

Sangwoo Shin

Ph.D. Student in Computer Science, University of Wisconsin–Madison
sshin237@wisc.edu | Website

RESEARCH INTERESTS

Robot Learning, Reinforcement Learning, Sim-to-Real Transfer, System Identification, Embodied AI

EDUCATION

University of Wisconsin–Madison
Ph.D. in Computer Science
Advisor: Professor Josiah P. Hanna

Madison, WI
Aug. 2025 – Present

Sungkyunkwan University
M.S. in Computer Science
Advisor: Professor Honguk Woo

Suwon, South Korea
Mar. 2022 – Aug. 2024

Sungkyunkwan University
B.S. in Mathematics, *Summa Cum Laude* (GPA: 3.83 / 4.00)

Suwon, South Korea
Mar. 2015 – Mar. 2022

PUBLICATIONS

*Note: Publications under the name Sangwoo Cho are mine prior to a name change. * denotes equal contribution.*

- Informationally Decoupled Trajectory Design for Sim-to-Real System Identification**
Sangwoo Shin, Ashvin Anilkumar, Ryan Gao, Josiah P. Hanna
International Conference on Robotics and Automation (ICRA), 2027, *in submission*
- RoboFQE: Rollout-Free Evaluation of Visual Robot Policies from Offline Trajectories**
Yunfu Deng, Sangwoo Shin, Brahma S. Pavse, Josiah P. Hanna
Conference on Robot Learning (CoRL), 2026
- Articulated-Body Dynamics Network: Dynamics-Grounded Prior for Robot Learning**
Sangwoo Shin, Kunzhao Ren, Xiaobin Xiong, Josiah P. Hanna
International Conference on Intelligent Robots and Systems (IROS), 2026
- LLM-Based Offline Learning for Embodied Agents via Consistency-Guided Reward Ensemble**
Yujeong Lee*, Sangwoo Shin*, Wei-Jin Park, Honguk Woo
Empirical Methods in Natural Language Processing (EMNLP) Findings, 2024
- Semantic Skill Grounding for Embodied Instruction-Following in Cross-Domain Environments**
Sangwoo Shin*, SeungHyun Kim*, Youngsoo Jang, Moontae Lee, Honguk Woo
Association for Computational Linguistics (ACL) Findings, 2024
- SemTra: A Semantic Skill Translator for Cross-domain Zero-shot Policy Adaptation**
Sangwoo Shin, Minjong Yoo, Jeongwoo Lee, Honguk Woo
AAAI Conference on Artificial Intelligence (AAAI), 2024
- One-shot Imitation in a Non-Stationary Environment via Multi-Modal Skill**
Sangwoo Shin, Dahee Lee, Minjong Yoo, Woo Kyung Kim, Honguk Woo
International Conference on Machine Learning (ICML), 2023
- Skills Regularized Task Decomposition for Multi-task Offline Reinforcement Learning**
Minjong Yoo, Sangwoo Cho, Honguk Woo
Neural Information Processing Systems (NeurIPS), 2022, *Oral Presentation*
- An Efficient Combinatorial Optimization Model Using Learning-to-Rank Distillation**

Hyunsung Lee*, **Sangwoo Cho***, Honguk Woo

*AAAI Conference on Artificial Intelligence (AAAI), 2022, **Oral Presentation***

10. **Differentiable Ranking Metric Using Relaxed Sorting for Top-K Recommendation**

Hyunsung Lee, **Sangwoo Cho**, Yeongjae Jang, Jaekwang Kim, Honguk Woo

IEEE Access, 2021

11. **A Global DAG Task Scheduler Using Deep Reinforcement Learning and Graph Convolution Network**

Hyunsung Lee, **Sangwoo Cho**, Yeongjae Jang, Jinkyu Lee, Honguk Woo

IEEE Access, 2021

RESEARCH & INDUSTRY EXPERIENCE

Diden Robotics — Robotics Intern

Dec. 2024 – May 2025

Advisors: Dr. Yong Um, Dr. Joon-Ha Kim

- Developed reinforcement learning frameworks for quadruped robots.

LG AI Research — Research Intern

Oct. 2023 – Jan. 2024

Advisors: Dr. Moontae Lee, Dr. Youngsoo Jang

- Researched controllable large language models using multi-objective reinforcement learning from human feedback.

OPEN-SOURCE CONTRIBUTIONS

NVIDIA Newton — Physics Engine

- Fixed a joint-indexing bug in Warp-to-MuJoCo coordinate conversion kernels and added regression tests (PR #2332).
- Fixed namespaced-attribute handling in `State.assign` via automatic `AttributeNamespace` discovery (PR #2487).

MJLAB — MuJoCo-based Reinforcement Learning Framework

- Fixed name/ID ordering mismatch in scene entity resolution (PR #876).

Genesis — Physics Engine

- Added `filter_link_idx` to contact sensors for counterpart-link filtering (PR #2772).
- Proposed `history_length` support for transient-contact detection (Issue #2646, PR #2655).

TEACHING

Teaching Assistant — University of Wisconsin–Madison

Aug. 2025 – Present

- CS 540: Introduction to Artificial Intelligence
- CS 320: Data Science Programming

Tutoring — **Linear Algebra**

Jan. 2020 – Jun. 2020

- Topics included module theory, Jordan canonical form, diagonalization, duality, and similarity of operators.

Tutoring — **Algebraic Topology**

Jan. 2021 – Jun. 2021

- Topics included fundamental groups, homotopy, and homology.